UMassAmherst
Manning College of Information & Computer Sciences

Correlating Accuracy in Importance Sampling with Human Screening Efforts

Huu Duy Nguyen, Haluk Yuzukirmizi, Egemen Dag, Muhammad Bilal

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MOTIVATION & BACKGROUND

- Counting objects in large images using computer vision demands balancing automation and human effort for efficient and accurate estimation.
- **DISCount** integrates **importance sampling** with computer vision, selectively vetting outputs to reduce exhaustive labeling (Fig. 1) while preserving accuracy level.
- However, there is little resource about the minimal manual efforts required for a desirable accuracy.

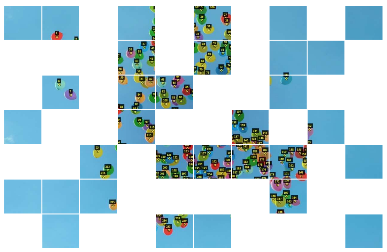

Fig. 1: Labeled and Tiled Data

PROBLEM STATEMENT

- This research explores whether object counting models require significant human effort using DISCount estimation.
- If human input is proved necessary, we then question how object counting models can **optimize** the trade-off between **automation and human labeling** (Fig. 2).

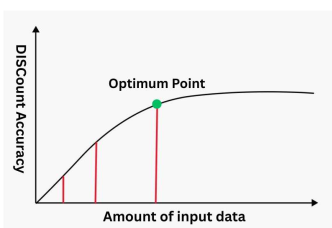

Fig. 2: Balancing and Plotting

METHODOLOGY

- In this study, we evaluated **DISCount (importance sampling)** accuracy by comparing the predicted count with **Monte Carlo (uniform) sampling** and **triangle sampling** (Fig. 3).

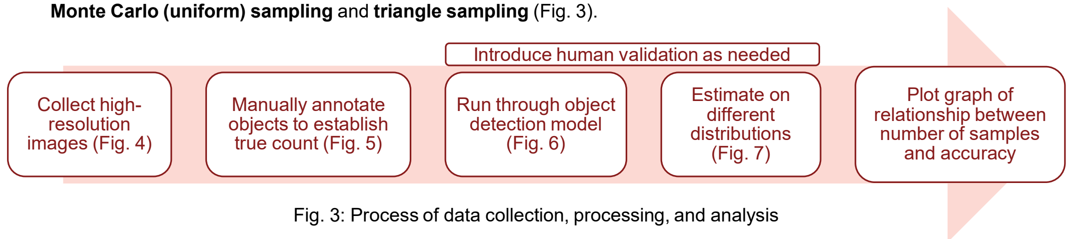

Fig. 3: Process of data collection, processing, and analysis

Fig. 4: Image from data set

Fig. 5: Manual labeling with bounding boxes

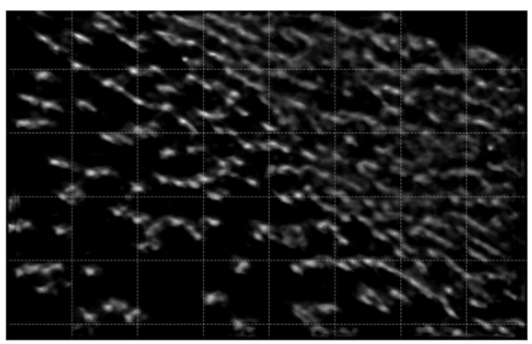

Fig. 6: Output Array from Computer Vision Model Visualized by Matplotlib

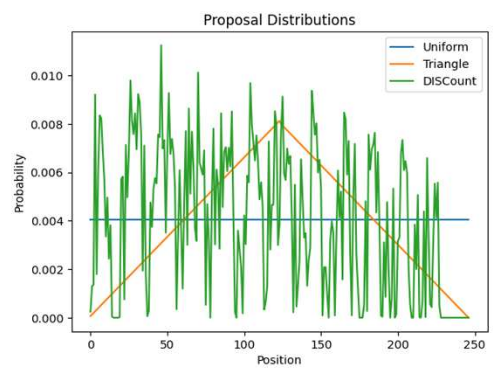

Fig. 7: Different distribution proposals

RESULT

- DISCount consistently achieved over 90% accuracy across multiple runs sampling only 20% of the tiles and kept the accuracy stably close to the true count (Fig. 8).
- Mathematically, because sampling is proportional to object density, manual correction strategies had **minimal impact** on overall accuracy.
- In all trials, human involvement subsided to **light supervising**, assuring the estimation is approaching the ground truth.

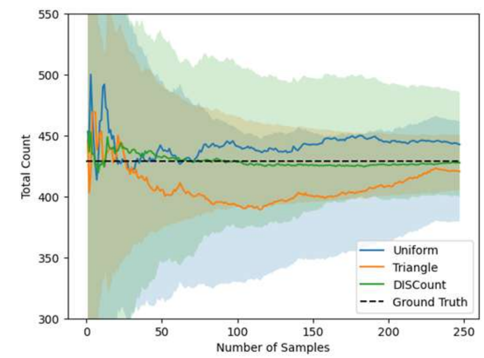

Fig. 8: Average count estimations of crowd image (Fig. 4) over 5 simulations using Uniform Sampling, Triangle Sampling, and Importance Sampling

FUTURE WORK

- DISCount can be run with low accuracy models, or with photos with great distort and noises, to determine if human screening is significant enough to measure correlation with accuracy.

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